



Random Average Sampling in a Reproducing Kernel Subspace of Mixed Lebesgue Space $L^{p,q}(\mathbb{R}^{n+1})$

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Abstract. In this paper, we study average sampling inequality in a probabilistic framework for a reproducing kernel subspace V of mixed Lebesgue space. More precisely, we show with high probability that a function concentrated on a compact cube C can be stably recovered from their $\mathcal{O}(\mu(C) \log \mu(C))$ many average values at uniformly distributed random points over C , where μ is a Lebesgue measure. Further, we propose an exponential convergence reconstruction scheme to reconstruct the concentrated function from their random average measurements and illustrate with an example.

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1. Introduction

The main goal of the sampling problem is to find a discrete sample set X under which a function f can be recovered from its samples $\{f(x_j) : x_j \in X\}$ without losing any information and vice versa. The classical Whittaker-Shannon-Kotel'nikov sampling theorem states that the function f in Paley-Wiener space $PW_{[-\frac{1}{2}, \frac{1}{2}]}(\mathbb{R})$, i.e., f in $L^2(\mathbb{R})$ and Fourier transform of f supported on a compact set $[-\frac{1}{2}, \frac{1}{2}]$, can be recovered from its uniform sample on $\mathbb{Z}$, and the reconstruction formula is given by

$$f(x) = \sum_{k \in \mathbb{Z}} f(k) \frac{\sin \pi(x-k)}{\pi(x-k)}.$$

It plays an important role in digital communication to convert a continuous voice signal into a discrete signal. However, in many applications, we deal with time-limited signals which are not band-limited due to the uncertainty principle. Consequently, many researchers considered the model space shift-invariant space for the sampling problem [1, 2, 18]. Later, Q. Sun and M. Nashed considered the sampling problem for the image space of an idempotent integral operator [30, 31], which generalizes the shift-invariant space and is more realistic for modeling signals with smoother spectrum. Over the last decade, the sampling problem has gradually been studied in the context of mixed Lebesgue space.

Mixed Lebesgue space $L^{p,q}(\mathbb{R}^{n+1})$ is a collection of measurable function f on $\mathbb{R}^n \times \mathbb{R}$ with

$$\|f\|_{L^{p,q}(\mathbb{R}^{n+1})}^q = \int_{\mathbb{R}} \left(\int_{\mathbb{R}^n} |f(x,y)|^p dx \right)^{\frac{q}{p}} dy < \infty, \text{ for } 1 \leq p, q < \infty.$$

The study of mixed Lebesgue space can be traced back to the 1960s [10, 20]. Later, it was intensively studied in different areas like operator theory [19], harmonic analysis [12], and many more. In investigations of the solutions of partial differential equations containing space-time variables, such as the heat or wave equation, mixed Lebesgue spaces are encountered, notably in Strichartz estimates; see, for example, [4, 24, 40].

Mixed Lebesgue space $L^{p,q}(\mathbb{R}^{n+1})$ is ideal for studying the sampling problem for the signal defined in the space and time domain. For example, a black and white video signal can be represented by a function $f(x, y, t)$, where $f(\cdot, \cdot, t)$ denotes the frame at time t , and $f(x, y, \cdot)$ denotes the brightness at point (x, y) . In mixed Lebesgue space, Torres and Ward investigated the uniform sampling problem for band-limited functions [40, 43]. After that, the sampling theorem was generalized in the case of non-uniform samples for shift-invariant space [27, 46, 47] and image space of an idempotent integral operator on $L^{p,q}(\mathbb{R}^{n+1})$ [23, 25]. However, in most physical cases, signal measuring devices are non-ideal to measure the sample value of the signal exactly at the sampling position. Therefore, a weighted average sample value of the signal is assumed in the neighborhood of the sampling point. In other words, for a function f on Ω and a sampling set $\Gamma \subset \Omega$, the measurement of average sampling data of f in the neighborhood of $\gamma \in \Gamma$ is of the form

$$\langle f, \psi_\gamma \rangle = \int_{\Omega} f(x) \psi_\gamma(x) dx,$$

where for each $\gamma \in \Gamma$, $\int_{\Omega} \psi_\gamma(x) dx = 1$ and ψ_γ supported in the neighborhood of γ . We call the set $\Psi = \{\psi_\gamma : \gamma \in \Gamma\}$ *localized average sampler*. In particular, the exact sample of f can be realized as average sample of f with respect to Dirac delta.

Mathematically the average sampling problem can be stated as follows: For a given function space $V \subseteq L^{p,q}(\mathbb{R}^{n+1})$, any $f \in V$ can be stably recovered from its average value $\{\langle f, \psi_{x_i, y_j} \rangle : i \in \mathbb{Z}^n, j \in \mathbb{Z}\}$ if there exist $A, B > 0$ such that

$$\begin{aligned} A\|f\|_{L^{p,q}(\mathbb{R}^{n+1})} &\leq \left(\sum_{j \in \mathbb{Z}} \left(\sum_{i \in \mathbb{Z}^n} |\langle f, \psi_{x_i, y_j} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \\ &\leq B\|f\|_{L^{p,q}(\mathbb{R}^{n+1})} \quad \forall f \in V. \end{aligned} \quad (1.1)$$

In that case, the sample set $X = \{(x_i, y_j) \in \mathbb{R}^{n+1} : i \in \mathbb{Z}^n, j \in \mathbb{Z}\}$ is called a *stable sample set*, and $\Psi = \{\psi_{x_i, y_j} : i \in \mathbb{Z}^n, j \in \mathbb{Z}\}$ is called a *stable sampler* for V . Nashed et al. [32] showed that any localized average sampler could not be a stable sampler for $L^2(\mathbb{R})$. Therefore, the average sampling problem is investigated in some subspaces of Lebesgue space. Gröchenig [17] first introduced the reconstruction of band-limited functions from their local average. Later, the average sampling problem was studied in various function subspaces of $L^p(\mathbb{R}^n)$ [3, 26, 37, 39], and mixed Lebesgue space $L^{p,q}(\mathbb{R}^{n+1})$ [21, 25, 42].

The random average sampling problem asks for the probability of stable recovery of signal from its average values at random points. In particular, the validation of the average sampling inequality (1.1) for randomly distributed points $X = \{(x_i, y_j) : i \in \mathbb{Z}^n, j \in \mathbb{Z}\}$. Bass and Gröchenig [8] showed that stable recovery of band-limited function from their exact sample at randomly distributed points over $\mathbb{R}^n$ is not possible. They considered a uniform distribution over each disjoint cube of $\mathbb{R}^n$ and spatial Poisson process over $\mathbb{R}^n$, and showed that with probability one, random samples are concentrated near zeros of the function and have an arbitrarily large gap. As a result, one can construct a class of functions with prescribed zeros and a class of functions with energy concentrated on those gaps, which fails to satisfy sampling inequality (1.1). Hence to get a positive result, Bass and Gröchenig considered a class of functions concentrated on compact domain $C_R = [-\frac{R}{2}, \frac{R}{2}]^n$, and uniform probability distribution of random samples over C_R . In particular, they proved the following theorem.

Theorem 1.1 ([8]). Let $\{x_j : j \in \mathbb{N}\}$ be a sequence of i.i.d. random variables uniformly distributed over C_R and $0 < \sigma < 1 - \delta$. Then the sampling inequality

$$\frac{1}{R^n}(1 - \delta - \sigma)\|f\|_{L^2(\mathbb{R}^n)} \leq \frac{1}{m} \sum_{j=1}^m |f(x_j)|^2 \leq \frac{1}{R^n}(1 + \sigma)\|f\|_{L^2(\mathbb{R}^n)} \quad \forall f \in PW(R, \delta) \quad (1.2)$$

holds with probability $1 - 2A \exp\left(-B \frac{m\sigma^2}{R^n(1+\sigma)}\right)$, for some $A, B > 0$. Here $PW(R, \delta)$ denotes the set $\{f \in PW_{[-\frac{1}{2}, \frac{1}{2}]^n}(\mathbb{R}^n) : \|f\|_{L^2(C_R^c)} < \delta\|f\|_{L^2(\mathbb{R}^n)}\}$.

Recovering signals from their random measurement is a well-known method in machine learning [13, 36] and compressed sensing [11, 14]. In the last decade, many authors studied the random sampling inequality (1.2) for different classes of function spaces. For example, multivariate trigonometric polynomials [7], Paley-Wiener space [8, 9], shift-invariant space [15, 44, 45], image of an idempotent integral operator on $L^p(\mathbb{R}^n)$ [29, 33, 34], and spherical reproducing kernel Hilbert space [35]. However, the study of random sampling in mixed-normed space is challenging due to the non-commutative nature of integral norm and discrete norm. In recent years, the random sampling problem was studied in a subspace of mixed Lebesgue space $L^{p,q}(\mathbb{R}^{n+1})$, such as shift-invariant space [5, 22, 41], and image space of an idempotent integral operator on $L^{p,q}(\mathbb{R}^{n+1})$ [16].

To the best of our knowledge, the random average sampling problem has not been investigated. In this paper, we study random average sampling and reconstruction of functions in image space V of an idempotent integral operator on $L^{p,q}(\mathbb{R}^{n+1})$. In connection to the set concentration function defined in Theorem 1.1, we consider a compact domain $C_{R,S} = [-\frac{R}{2}, \frac{R}{2}]^n \times [-\frac{S}{2}, \frac{S}{2}]$ with $R, S > 1$, and the set of δ -concentrated functions on $C_{R,S}$ in $V \subset L^{p,q}(\mathbb{R}^{n+1})$ is defined as follows

$$V^*(R, S, \delta) := \{f \in V : (1 - \delta) \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} \leq \|f\|_{L^{p,q}(C_{R,S})}\}.$$

We prove that the random sample set $X = \{(x_i, y_j) \in \mathbb{R}^n \times \mathbb{R} : i, j \in \mathbb{N}\}$ uniformly distributed over compact domain $C_{R,S}$ satisfies the average sampling inequality

$$\begin{aligned} A \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} &\leq \left(\frac{1}{m} \sum_{j=1}^m \left(\frac{1}{\ell} \sum_{i=1}^{\ell} |\langle f, \psi_{x_i, y_j} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \\ &\leq B \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}, \end{aligned} \quad (1.3)$$

for $V^*(R, S, \delta)$ with high probability.

The main investigations of the paper are outlined in the following exposition.

- (i) The random sampling problem in the framework of mixed norm was proved under the assumption of the strong decay condition of the reproducing kernel [5, 16] or the local approximation of the function space by the finite-dimensional subspace [28]. Following the idea of the result for the reproducing kernel subspace of $L^p(\mathbb{R}^n)$ [29], we drop the decay condition of the reproducing kernel and prove the random average sampling inequality (1.3).
- (ii) The random sampling problem is more efficient and prominent in practice if the random sample set of small size acquires the expected probability. In the case of reproducing kernel subspace V of $L^p(\mathbb{R}^n)$, it was proved for the set of concentrated functions on C_R that the sample size required to be at least of $\mathcal{O}(R^{2n})$ [8, 33, 44, 45], and later, it was improved

to $\mathcal{O}(R^n(\log R^n)^3)$ [34] and $\mathcal{O}(R^n \log R^n)$ [9, 15, 29]. However, due to the non-commutativity of the mixed norm, the random sampling problem in a reproducing kernel subspace V of $L^{p,q}(\mathbb{R}^{n+1})$ leads to the conclusion that the effective number of samples is indeed $\mathcal{O}(R^{(n+1)(2p+1)})$, in case of $1 < p = q < \infty$ and the set of concentrated functions on $C_{R,R}$, see [5, 16, 22, 41]. The recent article by Y. Li [28] has effectively established the solution to the random sampling problem within the context of reproducing kernel subspace of mixed Lebesgue space. However, it does not provide an explicit relation between the order of the sample size and the corresponding probability bound. In this article, we show that the function concentrated on $C_{R,S}$ can be reconstructed from its average value taken at uniformly distributed random sample set with high probability if the sample size is at least $\mathcal{O}(R^n S \log(R^n S))$.

- (iii) Reconstruction of a function from its samples is another major problem in sampling theory. There are well-known frame theoretic method, linear algebraic method, and iterative frame method to reconstruct the original function from its stable sample. As the set $V^*(R, S, \delta)$ is not a subspace, these techniques are no longer applicable. However, using the fact that the normalized set $V^*(R, S, \delta)$ is totally bounded with respect to $\|\cdot\|_{L^\infty, \infty(C_{R,S})}$, we can approximate the function on $C_{R,S}$, see [5, 22]. At the same time, it does not explain the behavior of the approximated function on $\mathbb{R}^n \setminus C_{R,S}$, and it raises the following questions: How much is the approximated function concentrated on $C_{R,S}$? Is it a good approximation on $\mathbb{R}^{n+1}$? In this paper, we propose an iterative reconstruction algorithm to reconstruct the concentrated function from their random average samples on $C_{R,S}$ and clarify those questions.

The rest of the paper is organized as follows. In Sect. 2, we present some notations and definitions. We also give some preliminary results and assumptions. Section 3 is devoted to studying random average sampling inequality for the concentrated functions. We discuss the reconstruction formula to recover the concentrated function from their random average samples and give a numerical illustration of the reconstruction algorithm in Sect. 4.

2. Preliminaries

In this section, we recall some basic definitions of function sets. We define reproducing kernel space V and give some preliminary results.

The mixed normed space $\ell^{p,q}(\mathbb{Z}^{n+1})$ is a collection of all sequences $\{c = c(i, j) : i \in \mathbb{Z}^n, j \in \mathbb{Z}\}$ such that

$$\|c\|_{\ell^{p,q}}^q = \sum_{j \in \mathbb{Z}} \left(\sum_{i \in \mathbb{Z}^n} |c(i, j)|^p \right)^{\frac{q}{p}} < \infty.$$

It is easy to verify that $\ell^{\infty,\infty}(\mathbb{Z}^{n+1}) = \ell^\infty(\mathbb{Z}^{n+1})$, in fact $L^{\infty,\infty}(\mathbb{R}^{n+1}) = L^\infty(\mathbb{R}^{n+1})$.

Our model space V is an image space of an idempotent integral operator, where the integral operator $T : L^{p,q}(\mathbb{R}^{n+1}) \rightarrow L^{p,q}(\mathbb{R}^{n+1})$ is defined as

$$Tf(x, y) = \int_{\mathbb{R}} \int_{\mathbb{R}^n} K(x, y, s, t) f(s, t) ds dt, \quad (2.1)$$

and $T^2 = T$. In addition, we assume that for some $0 < \theta \leq 1$, the integral kernel K satisfies

$$\begin{aligned} \kappa_\theta := & \|(x, s) \mapsto \|K(x, \cdot, s, \cdot)\|_W\|_W \\ & + \sup_{0 \leq \varepsilon < 1} \varepsilon^{-\theta} \|(x, s) \mapsto \|w_\varepsilon(K)(x, \cdot, s, \cdot)\|_W\|_W < \infty, \end{aligned} \quad (2.2)$$

where $w_\varepsilon(K)(x, y, s, t) = \sup_{(x', y') \in B_\varepsilon(x, y)(s', t') \in B_\varepsilon(s, t)} |K(x', y', s', t') - K(x, y, s, t)|$, $B_r(a, b)$ is an open ball of radius r and center at (a, b) with respect to $\|\cdot\|_\infty$ in $\mathbb{R}^{n+1}$, and for a function K_0 defined on $\mathbb{R}^n \times \mathbb{R}^n$, we denote

$$\|K_0\|_W := \max \left\{ \sup_{x \in \mathbb{R}^n} \|K_0(x, \cdot)\|_{L^1(\mathbb{R}^n)}, \sup_{y \in \mathbb{R}^n} \|K_0(\cdot, y)\|_{L^1(\mathbb{R}^n)} \right\}.$$

Remark 2.1. A prerequisite of strong decay condition together with the regularity of the reproducing kernel was commonly assumed in [5, 16, 23]. The decay assumption subsequently leads to the off-diagonal decay condition, i.e., $\|(x, s) \mapsto \|K(x, \cdot, s, \cdot)\|_W\|_W < \infty$. The off-diagonal decay of the kernel K implies that the integral operator (2.1) is bounded. The regularity condition

$$\lim_{\varepsilon \rightarrow 0} \|(x, s) \mapsto \|w_\varepsilon(K)(x, \cdot, s, \cdot)\|_W\|_W = 0$$

ensures that the image space of the idempotent integral is a reproducing kernel space. The inequality (2.2) implies that the integral kernel satisfies the off-diagonal decay condition and $\|(x, s) \mapsto \|w_\varepsilon(K)(x, \cdot, s, \cdot)\|_W\|_W$ converges to 0 with the rate of ε^θ .

A closed subspace V of $L^{p,q}(\mathbb{R}^{n+1})$ is a reproducing kernel space if every point evaluation map $f \mapsto f(x, y)$, $(x, y) \in \mathbb{R}^{n+1}$ is continuous on V , i.e., for each $x \in \mathbb{R}^{n+1}$ there exists $C_x > 0$ such that

$$|f(x)| \leq C_x \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}, \quad \forall f \in V.$$

Theorem 2.2 ([23]). *If the integral kernel K defined in (2.1) satisfies off-diagonal decay and Hölder continuity (2.2), then the following are true:*

- (i) *The space $V := \text{Range}(T) = \{Tf : f \in L^{p,q}(\mathbb{R}^{n+1})\} = \{f \in L^{p,q}(\mathbb{R}^{n+1}) : Tf = f\}$ is a reproducing kernel subspace of $L^{p,q}(\mathbb{R}^{n+1})$.*
- (ii) *For any $(s, t) \in \mathbb{R}^{n+1}$, $K(\cdot, \cdot, s, t) \in V$.*

Example 2.3. The mixed Wiener amalgam space $W(L^{p,q})(\mathbb{R}^{n+1})$ is a set of all measurable functions f such that

$$\|f\|_{W(L^{p,q})(\mathbb{R}^{n+1})} := \left(\sum_{j \in \mathbb{Z}} \sup_{y \in [0,1]} \left(\sum_{i \in \mathbb{Z}^n} \sup_{x \in [0,1]^n} |f(x+i, y+j)|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} < \infty.$$

Let $0 < m \leq M < \infty$ and $\varphi \in W(L^{1,1})(\mathbb{R}^{n+1})$ be a Hölder continuous function satisfying

$$m\|c\|_{\ell^2(\mathbb{Z}^{n+1})} \leq \left\| \sum_{k \in \mathbb{Z}^{n+1}} c_k \varphi(\cdot - k) \right\|_{L^2(\mathbb{R}^{n+1})} \leq M\|c\|_{\ell^2(\mathbb{Z}^{n+1})} \quad \forall c \in \ell^2(\mathbb{Z}^{n+1}).$$

Then there exists $\tilde{\varphi} \in W(L^{1,1})(\mathbb{R}^{n+1})$ such that $\langle \varphi(\cdot - i, \cdot - j), \tilde{\varphi}(\cdot - i', \cdot - j') \rangle = \delta_{(i,j)}^{(i',j')}$, where $\delta_{(i,j)}^{(i',j')}$ stands for the Kronecker symbol, see [27]. Further, for $\varepsilon \in (0, 1)$ and $1 \leq p, q < \infty$, if $\|w_\varepsilon(\varphi)\|_{W(L^{1,1})(\mathbb{R}^{n+1})} < \varepsilon$ then the shift-invariant space

$$V_{p,q}(\varphi) := \left\{ \sum_{j \in \mathbb{Z}} \sum_{i \in \mathbb{Z}^n} c_{i,j} \varphi(\cdot - i, \cdot - j) : c = (c_{i,j}) \in \ell^{p,q}(\mathbb{Z}^{n+1}) \right\}$$

is an image of an idempotent integral operator with the kernel K is given by

$$K(x, y, s, t) = \sum_{j \in \mathbb{Z}} \sum_{i \in \mathbb{Z}^n} \varphi(x - i, y - j) \tilde{\varphi}(s - i, t - j),$$

and the integral kernel K satisfies (2.2) with $\theta = 1$.

Example 2.4. Let the set $\Gamma \subset \mathbb{R}^{n+1}$ be relatively separated with a positive gap σ , i.e.,

$$A_\Gamma(\sigma) = \sup_{(x,y) \in \mathbb{R}^{n+1}} \sum_{\gamma \in \Gamma} \chi_{B_\sigma(\gamma)}(x, y) < \infty, \text{ and} \\ B_\Gamma(\sigma) = \inf_{(x,y) \in \mathbb{R}^{n+1}} \sum_{\gamma \in \Gamma} \chi_{B_\sigma(\gamma)}(x, y) > 1.$$

Let $\Phi = \{\phi_\gamma\}_{\gamma \in \Gamma} \subset L^{p,q}(\mathbb{R}^{n+1})$ and $\tilde{\Phi} = \{\tilde{\phi}_\gamma\}_{\gamma \in \Gamma} \subset L^{p',q'}(\mathbb{R}^{n+1})$ be two families of functions such that

$$\int_{\mathbb{R}} \int_{\mathbb{R}^n} \phi_\gamma(x, y) \tilde{\phi}_{\gamma'}(x, y) dx dy = \delta_{\gamma'}^{\gamma}.$$

Then the integral operator T in (2.1) with the kernel $K(x, y, s, t) = \sum_{\gamma \in \Gamma} \phi_\gamma(x, y) \tilde{\phi}_\gamma(s, t)$ is an idempotent operator on $L^{p,q}(\mathbb{R}^{n+1})$ [23], and the corresponding reproducing kernel space is

$$V(\Phi) := \left\{ \sum_{\gamma \in \Gamma} c_\gamma \phi_\gamma : (c_\gamma) \in \ell^{p,q}(\Gamma) \right\}.$$

We further assume that there exists $h \in W(L^{1,1})(\mathbb{R}^{n+1})$ and for $\varepsilon \in (0, 1)$ there exists $g \in W(L^{1,1})(\mathbb{R}^{n+1})$ with $\|g\|_{W(L^{1,1})(\mathbb{R}^{n+1})} < \varepsilon$ such that

$$|\phi_\gamma(z)| + |\tilde{\phi}_\gamma(z)| \leq h(z - \gamma),$$

$$\text{and } |w_\varepsilon(\phi_\gamma)(z)| + |w_\varepsilon(\tilde{\phi}_\gamma)(z)| \leq g(z - \gamma),$$

for all $\gamma \in \Gamma$ and $z \in \mathbb{R}^{n+1}$. Then one can verify that the kernel K satisfies (2.2) with $\theta = 1$.

Definition 2.5. Let $X = \{(x_i, y_j) : i, j \in \mathbb{N}\}$ be a countable subset of $C_{R,S}$. A positive constant $\varepsilon_X(\ell, m)$ is called *covering radius* for $C_{R,S}$ if

$$\varepsilon_X(\ell, m) := \inf \left\{ r > 0 : C_{R,S} \subseteq \bigcup_{j=1}^m \bigcup_{i=1}^\ell B_r(x_i, y_j) \right\}.$$

In particular, for a given finite collection $X = \{(x_i, y_j) : 1 \leq i \leq \ell, 1 \leq j \leq m\} \subset C_{R,S}$ with covering radius $\varepsilon = \varepsilon_X(\ell, m)$, there exists a collection of functions $\{\beta_{ij} : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ such that

$$(i) \ 0 \leq \beta_{ij} \leq 1, \quad (ii) \ \text{supp}(\beta_{ij}) \subseteq B_\varepsilon(x_i, y_j),$$

$$(iii) \ \sum_{j=1}^m \sum_{i=1}^\ell \beta_{ij}(x, y) = 1, \quad \forall (x, y) \in C_{R,S}.$$

The collection of functions $\{\beta_{ij}\}$ is called partition of unity of $C_{R,S}$ associated to the open cover $\{B_\varepsilon(x_i, y_j) : (x_i, y_j) \in X\}$.

Definition 2.6. Let $X = \{(x_i, y_j) : i, j \in \mathbb{N}\}$ be a set of points in $\mathbb{R}^{n+1}$. For $\eta > 0$ and $f \in V$, the average sampling value of f in the neighborhood of (x_i, y_j) is defined as follows:

$$\langle f, \psi_{x_i, y_j} \rangle = \int_{\mathbb{R}} \int_{\mathbb{R}^n} f(x, y) \psi_{x_i, y_j}(x, y) dx dy,$$

where $\{\psi_{x_i, y_j}\}$ is a collection of averaging functions satisfying following conditions:

For each $(x_i, y_j) \in X$

- (i) $\text{supp}(\psi_{x_i, y_j}) \subset B_\eta(x_i, y_j)$,
- (ii) $\int_{\mathbb{R}^{n+1}} \psi_{x_i, y_j}(x, y) dx dy = 1$,
- (iii) there exists $M > 0$ such that $|\psi_{x_i, y_j}(x, y)| \leq M$ for all $(x, y) \in \mathbb{R}^{n+1}$.

It is known that given a sample set with sufficiently small gap, the signal may not be recovered from their local average values when η is large (e.g. [38, Theorem 2.3]). Hence, in this framework, we consider $0 < \eta < \min \left\{ 1, \left(\frac{1-\delta}{2M\kappa_\theta} \right)^{\frac{1}{n+1}} \right\}$.

3. Random Average Sampling of δ -Concentrated Signals

Average sampling problem with the mixed norm for an image of an idempotent integral operator studied in [23, 25]. If the gap of a relatively separated sample set is small enough, then functions are reconstructed stably and uniquely from the sample set. In the following theorem, we obtain the condition on the sample set (in terms of covering radius) to be a stable sampling set for the set of δ -concentrated functions.

Theorem 3.1. *Let $1 \leq p, q < \infty$, $X = \{(x_i, y_j) : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ be a sequence of points in $C_{R,S}$, and $\{\psi_{x_i, y_j} : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ be a collection of averaging functions with $\text{supp}(\psi_{x_i, y_j}) \subset B_\eta(x_i, y_j)$. Suppose the covering radius $\varepsilon = \varepsilon_X(\ell, m)$ is less than $(\frac{1-\delta}{M\kappa_\theta\eta^{n+1}})^{\frac{1}{\theta}} - \eta$, then the average sampling inequality*

$$\begin{aligned} & \frac{c_1}{\ell^{\frac{1}{p}} m^{\frac{1}{q}} R^{\frac{n}{p}} S^{\frac{1}{q}}} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} \\ & \leq \left(\frac{1}{m} \sum_{j=1}^m \left(\frac{1}{\ell} \sum_{i=1}^{\ell} |\langle f, \psi_{x_i, y_j} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \leq c_2 \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}. \end{aligned} \quad (3.1)$$

holds for all $f \in V^*(R, S, \delta)$. Here $c_1 = 1 - \delta - M\kappa_\theta\eta^{n+1}(\varepsilon + \eta)^\theta$ and $c_2 = M\eta^{\frac{n}{p'} + \frac{1}{q'}}$.

Proof. Let ε be the covering radius of X for $C_{R,S}$, and $f \in V^*(R, S, \delta)$ be arbitrary. The average value of f at (x_i, y_j) is denoted by $\langle f, \psi_{x_i, y_j} \rangle$. Then by Hölder's inequality we have

$$\begin{aligned} |\langle f, \psi_{x_i, y_j} \rangle| & \leq \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} \|\psi_{x_i, y_j}\|_{L^{p',q'}(\mathbb{R}^{n+1})} \\ & \leq M\eta^{\frac{n}{p'} + \frac{1}{q'}} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}, \\ \left(\sum_{j=1}^m \left(\sum_{i=1}^{\ell} |\langle f, \psi_{x_i, y_j} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} & \leq \ell^{\frac{1}{p}} m^{\frac{1}{q}} M\eta^{\frac{n}{p'} + \frac{1}{q'}} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}. \end{aligned}$$

To calculate the lower bound of the average sampling inequality, we define a bounded approximation operator

$$Af = \sum_{j=1}^m \sum_{i=1}^{\ell} \langle f, \psi_{x_i, y_j} \rangle \beta_{ij},$$

where $\{\beta_{ij}\}$ is a partition of unity of $C_{R,S}$ associated to the open cover $\{B_\varepsilon(x_i, y_j) : (x_i, y_j) \in X\}$. Then we have

$$\begin{aligned}
\|Af\|_{L^{p,q}(C_{R,S})} &= \left(\int_{-\frac{\ell}{2}}^{\frac{\ell}{2}} \left(\int_{[-\frac{R}{2}, \frac{R}{2}]^n} \left| \sum_{j=1}^m \sum_{i=1}^{\ell} \langle f, \psi_{x_i, y_j} \rangle \beta_{ij}(x, y) \right|^p dx \right)^{\frac{q}{p}} dy \right)^{\frac{1}{q}} \\
&\leq \left(\int_{-\frac{\ell}{2}}^{\frac{\ell}{2}} \left(\int_{[-\frac{R}{2}, \frac{R}{2}]^n} \|\{ \langle f, \psi_{x_i, y_j} \rangle\}\|_{\ell^{p,q}}^p \|\{\beta_{ij}(x, y)\}\|_{\ell^{p',q'}}^p dx \right)^{\frac{q}{p}} dy \right)^{\frac{1}{q}} \\
&\leq \|\{ \langle f, \psi_{x_i, y_j} \rangle\}\|_{\ell^{p,q}} \left(\int_{-\frac{\ell}{2}}^{\frac{\ell}{2}} \left(\int_{[-\frac{R}{2}, \frac{R}{2}]^n} 1 dx \right)^{\frac{q}{p}} dy \right)^{\frac{1}{q}} \\
&\leq \ell^{\frac{1}{p}} m^{\frac{1}{q}} R^{\frac{n}{p}} S^{\frac{1}{q}} \left(\frac{1}{m} \sum_{j=1}^m \left(\frac{1}{\ell} \sum_{i=1}^{\ell} |\langle f, \psi_{x_i, y_j} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}}.
\end{aligned}$$

Now for $(x, y) \in C_{R,S}$,

$$\begin{aligned}
f(x, y) - Af(x, y) &\leq \sum_{j=1}^m \sum_{i=1}^{\ell} (f(x, y) - \langle f, \psi_{x_i, y_j} \rangle) \beta_{ij}(x, y) \\
&\leq \sum_{j=1}^m \sum_{i=1}^{\ell} \beta_{ij}(x, y) \int_{\mathbb{R}^{n+1}} (f(x, y) - f(t)) \psi_{x_i, y_j}(t) dt \\
|f(x, y) - Af(x, y)| &\leq \sum_{j=1}^m \sum_{i=1}^{\ell} \beta_{ij}(x, y) \int_{\mathbb{R}^{n+1}} |f(x, y) - f(t)| |\psi_{x_i, y_j}(t)| dt \\
&\leq \sum_{j=1}^m \sum_{i=1}^{\ell} M\eta^{n+1} w_{\varepsilon+\eta}(f)(x, y) \beta_{ij}(x, y) \\
&= M\eta^{n+1} w_{\varepsilon+\eta}(f)(x, y),
\end{aligned}$$

where

$$\begin{aligned}
w_{\varepsilon+\eta}(f)(x, y) &= \sup_{\|(x', y') - (x, y)\|_{\infty} < \varepsilon + \eta} |f(x', y') - f(x, y)| \\
&\leq \sup_{\|(x', y') - (x, y)\|_{\infty} < \varepsilon + \eta} \int_{\mathbb{R}} \int_{\mathbb{R}^n} |K(x', y', s, t) \\
&\quad - K(x, y, s, t)| |f(s, t)| ds dt \\
&\leq \int_{\mathbb{R}} \int_{\mathbb{R}^n} w_{\varepsilon+\eta}(K)(x, y, s, t) |f(s, t)| ds dt.
\end{aligned}$$

Hence

$$\begin{aligned}
\|f - Af\|_{L^{p,q}(C_{R,S})} &\leq M\eta^{n+1} \|w_{\varepsilon+\eta}(f)\|_{L^{p,q}(C_{R,S})} \\
&\leq M\kappa_{\theta}\eta^{n+1} (\varepsilon + \eta)^{\theta} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}.
\end{aligned}$$

The operator A is bounded below for $V^*(R, S, \delta)$ on $C_{R,S}$ and the lower bound is given by

$$\begin{aligned}
\|Af\|_{L^{p,q}(C_{R,S})} &\geq \|f\|_{L^{p,q}(C_{R,S})} - \|f - Af\|_{L^{p,q}(C_{R,S})} \\
&\geq (1 - \delta) \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} - M\kappa_{\theta}\eta^{n+1} (\varepsilon + \eta)^{\theta} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}.
\end{aligned}$$

Therefore, for $f \in V^*(R, S, \delta)$

$$\frac{\left(1 - \delta - M\kappa_\theta \eta^{n+1}(\varepsilon + \eta)^\theta\right)}{\ell^{\frac{1}{p}} m^{\frac{1}{q}} R^{\frac{n}{p}} S^{\frac{1}{q}}} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} \leq \left(\frac{1}{m} \sum_{j=1}^m \left(\frac{1}{\ell} \sum_{i=1}^{\ell} |\langle f, \psi_{x_i, y_j} \rangle|^p\right)^{\frac{q}{p}}\right)^{\frac{1}{q}}.$$

This completes the proof. $\square$

Remark 3.2. The average sampling inequality (3.1) for the set of concentrated functions on $C_{R,S}$ can be simplified to the classical sampling inequality if the averaging function considered as the Dirac delta functional. For instance, the lower bound of the sampling inequality can be derived in a similar manner, while the upper bound is obtained through the application of the reproducing kernel property of V .

The above theorem implies that a finite collection X of samples in $C_{R,S}$ is a stable sampling set for δ -concentration functions when the covering radius of X bounded above by certain constant. Therefore, the random average sampling problem reduces to estimate the probability of the covering number

$$\varepsilon_X(\ell, m) > \sigma, \quad \text{where } 0 < \sigma < 1.$$

In particular, we prove the following theorem.

Theorem 3.3. Let $X = \{(x_i, y_j) : i, j \in \mathbb{N}\}$ be a sequence of i.i.d. random variable that are drawn uniformly from $C_{R,S}$ and $\{\psi_{x_i, y_j} : i, j \in \mathbb{N}\}$ be a collection of averaging functions with $\text{supp}(\psi_{x_i, y_j}) \subset B_\eta(x_i, y_j)$. For every $f \in V^*(R, S, \delta)$, the average sampling inequality

$$\begin{aligned} \frac{(1 - \delta)}{2\ell^{\frac{1}{p}} m^{\frac{1}{q}} R^{\frac{n}{p}} S^{\frac{1}{q}}} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} &\leq \left(\frac{1}{m} \sum_{j=1}^m \left(\frac{1}{\ell} \sum_{i=1}^{\ell} |\langle f, \psi_{x_i, y_j} \rangle|^p\right)^{\frac{q}{p}}\right)^{\frac{1}{q}} \\ &\leq M\eta^{\frac{n}{p} + \frac{1}{q}} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}, \end{aligned} \quad (3.2)$$

holds with probability at least

$$\begin{aligned} 1 - \left(\frac{4}{(2M\kappa_\theta \eta^{n+1})^{-\frac{1}{\theta}} (1 - \delta)^{\frac{1}{\theta}} - \eta}\right)^{n+1} \\ \times R^n S \exp\left(-\ell m \frac{((2M\kappa_\theta \eta^{n+1})^{-\frac{1}{\theta}} (1 - \delta)^{\frac{1}{\theta}} - \eta)^{n+1}}{2^{n+1} R^n S}\right). \end{aligned}$$

Proof. In order to calculate the probability of the average sampling inequality (3.2), we first compute the probability bound for the event $\mathcal{E} = \{\varepsilon_X(l, m) > \sigma\}$, where $0 < \sigma < 1$ and $\varepsilon_X(l, m)$ is a covering radius of i.i.d. random sequence of points $X = \{(x_i, y_j) : 1 \leq i \leq l, 1 \leq j \leq m\}$ uniformly distributed over $C_{R,S}$. The probability estimate of the event $\mathcal{E}$ was essentially proved in [7, 29], but for the sake of exposition, we shall provide a proof.

Let the event $\mathcal{E}$ be true and the collection $\Gamma \subseteq C_{R,S}$ be such that $\{B_{\frac{\sigma}{2}}(\gamma) : \gamma \in \Gamma\}$ covers the set $C_{R,S}$. Then there exists a point $\gamma_0 \in \Gamma$ such that $B_{\frac{\sigma}{2}}(\gamma_0) \cap X = \phi$. Therefore, the event $\mathcal{E}$ is a subset of the event $\mathcal{F} = \{B_{\frac{\sigma}{2}}(\gamma) \cap X = \phi \text{ for some } \gamma \in \Gamma\}$. There are infinitely many choices of such Γ , but in our computation, we consider Γ as a finite $\frac{\sigma}{2}$ -net of $C_{R,S}$, and the cardinality of Γ is bounded by

$$\#\Gamma \leq \left(\frac{2R}{\sigma} + 1\right)^n \left(\frac{2S}{\sigma} + 1\right).$$

The existence of such Γ is ensured by [13, Theorem 5.3]. Moreover, for any $\gamma \in \Gamma$ and $0 < r \leq 1$, there exists $z_\gamma \in B_r(\gamma)$, such that

$$B_{\frac{r}{2}}(z_\gamma) \subset B_r(\gamma) \cap C_{R,S}.$$

Therefore, $\mathbb{P}(\mathcal{E}) \leq \mathbb{P}(\mathcal{F})$ implies,

$$\begin{aligned} \mathbb{P}(\varepsilon_X(l, m) > \sigma) &\leq \mathbb{P}\left(B_{\frac{\sigma}{2}}(\gamma) \cap X = \phi \text{ for some } \gamma \in \Gamma\right) \\ &\leq \sum_{\gamma \in \Gamma} \mathbb{P}\left(B_{\frac{\sigma}{2}}(\gamma) \cap X = \phi\right) \\ &\leq \sum_{\gamma \in \Gamma} \left(1 - \frac{\mu(B_{\frac{\sigma}{2}}(\gamma) \cap C_{R,S})}{R^n S}\right)^{\ell m} \\ &\leq \sum_{\gamma \in \Gamma} \left(1 - \frac{\mu(B_{\frac{\sigma}{4}}(z_\gamma))}{R^n S}\right)^{\ell m} \\ &\leq \sum_{\gamma \in \Gamma} \left(1 - \frac{\sigma^{n+1}}{2^{n+1} R^n S}\right)^{\ell m} \\ &\leq \left(\frac{4}{\sigma}\right)^{n+1} R^n S \left(1 - \frac{\sigma^{n+1}}{2^{n+1} R^n S}\right)^{\ell m} \\ \mathbb{P}(\varepsilon_X(l, m) > \sigma) &\leq \left(\frac{4}{\sigma}\right)^{n+1} R^n S \exp\left(-\ell m \frac{\sigma^{n+1}}{2^{n+1} R^n S}\right). \end{aligned} \quad (3.3)$$

By Theorem 3.1, the average sampling inequality (3.1) hold for all $f \in V^*(R, S, \delta)$ if $\varepsilon_X(\ell, m) \leq \sigma$, where $\sigma = \kappa_\theta^{-\frac{1}{\theta}} \left(\frac{1-\delta}{2M\eta^{n+1}}\right)^{\frac{1}{\theta}} - \eta > 0$, as $\eta < \min\left\{1, \left(\frac{1-\delta}{2M\kappa_\theta}\right)^{\frac{1}{n+1}}\right\}$. In that case,

$$M\kappa_\theta \eta^{n+1} (\varepsilon + \eta)^\theta \leq \frac{1-\delta}{2}.$$

Hence, with probability $\mathbb{P}(\varepsilon_X(l, m) \leq \sigma)$, the average sampling inequality

$$\begin{aligned} \frac{(1-\delta)}{2\ell^{\frac{1}{p}}m^{\frac{1}{q}}R^{\frac{n}{p}}S^{\frac{1}{q}}}\|f\|_{L^{p,q}(\mathbb{R}^{n+1})} &\leq \left(\frac{1}{m}\sum_{j=1}^m\left(\frac{1}{\ell}\sum_{i=1}^{\ell}|\langle f, \psi_{x_i, y_j} \rangle|^p\right)^{\frac{q}{p}}\right)^{\frac{1}{q}} \\ &\leq M\eta^{\frac{n}{p}+\frac{1}{q}}\|f\|_{L^{p,q}(\mathbb{R}^{n+1})}. \end{aligned}$$

hold for all $f \in V^*(R, S, \delta)$. Therefore, the average sampling inequality (3.2) hold with probability at least

$$\begin{aligned} &1 - \left(\frac{4}{(2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta}\right)^{n+1} \\ &\times R^n S \exp\left(-\ell m \frac{((2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta)^{n+1}}{2^{n+1}R^n S}\right). \end{aligned}$$

This completes the proof. $\square$

Remark 3.4. For $\tau \in (0, 1)$ and $1 \leq p, q < \infty$, the random average sampling inequality (1.3) hold with probability at least $1 - \tau$, if

$$\begin{aligned} &\left(\frac{4}{(2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta}\right)^{n+1} \\ &\times R^n S \exp\left(-\ell m \frac{((2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta)^{n+1}}{2^{n+1}R^n S}\right) < \tau \\ &\exp\left(\ell m \frac{((2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta)^{n+1}}{2^{n+1}R^n S}\right) \\ &> \left(\frac{4}{(2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta}\right)^{n+1} \frac{R^n S}{\tau} \\ &\ell m \frac{((2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta)^{n+1}}{2^{n+1}R^n S} \\ &> \ln\left[\left(\frac{4}{(2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta}\right)^{n+1} \frac{R^n S}{\tau}\right] \\ &\ell m > \frac{2^{n+1}R^n S}{((2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta)^{n+1}} \\ &\times \ln\left[\left(\frac{4}{(2M\kappa_{\theta}\eta^{n+1})^{-\frac{1}{\theta}}(1-\delta)^{\frac{1}{\theta}}-\eta}\right)^{n+1} \frac{R^n S}{\tau}\right]. \end{aligned}$$

In particular, the random sampling inequality (1.3) hold with probability $1 - \tau$ if the sample size is at least $O(R^n S \ln(R^n S/\tau))$.

4. Reconstruction of a Function from Random Average Samples

The set $V^*(R, S, \delta)$ is a non-convex subset of linear space V , therefore the function reconstruction algorithm used in [2, 3, 17, 21, 25, 31, 42], cannot be directly applicable. Following the idea of Li et al. [29], we propose the approximation-projection iterative algorithm for the reconstruction of function in $V^*(R, S, \delta)$ from their average samples at random points in $C_{R,S}$.

Theorem 4.1. *Let T be an idempotent integral operator on $L^{p,q}(\mathbb{R}^{n+1})$ defined by (2.1) and the integral kernel K satisfies (2.2). Let $0 < \delta < \kappa_\theta^{-2}$, and $0 < \eta < \left(\frac{\kappa_\theta^{-2} - \delta}{M}\right)^{\frac{1}{n+1}}$. If $X = \{(x_i, y_j) : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ is a sequence of i.i.d. random variable that are drawn uniformly from $C_{R,S}$. Then $f \in V^*(R, S, \delta)$ can be uniquely reconstructed from its average sample $\{\langle f, \psi_{x_i, y_j} \rangle : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ by the following iterative algorithm:*

$$f_0 = S_X f, \quad f_k = f_0 + f_{k-1} - S_X f_{k-1}, \quad k \geq 1,$$

and

$$S_X f(x, y) = \sum_{j=1}^m \sum_{i=1}^{\ell} \langle f, \psi_{x_i, y_j} \rangle T\beta_{ij}(x, y),$$

with probability at least

$$1 - \left(\frac{4}{((\kappa_\theta^{-2} - \delta)/M\eta^{n+1})^{\frac{1}{\theta}} - \eta} \right)^{n+1} R^n S \exp \left(-\ell m \frac{\left(((\kappa_\theta^{-2} - \delta)/M\eta^{n+1})^{\frac{1}{\theta}} - \eta \right)^{n+1}}{2^{n+1} R^n S} \right),$$

where $\{\psi_{x_i, y_j} : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ and $\{\beta_{ij} : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ are the the averaging function and the partition of unity of $C_{R,S}$ respectively.

Moreover, f_k converges to f exponentially

$$\begin{aligned} \|f_k - f\|_{L^{p,q}(\mathbb{R}^{n+1})} &\leq \frac{1 + (M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2}{1 - (M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2} \\ &\quad \times \left((M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2 \right)^{k+1} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}, \end{aligned}$$

where ε is the covering radius for $C_{R,S}$ associated to the sample set X .

Proof. Let $X = \{(x_i, y_j) : 1 \leq i \leq \ell, 1 \leq j \leq m\}$ be a sequence of points in $C_{R,S}$ and $(x, y) \in \mathbb{R}^{n+1}$ then consider

$$S_X f(x, y) = \sum_{j=1}^m \sum_{i=1}^{\ell} \langle f, \psi_{x_i, y_j} \rangle T\beta_{ij}(x, y),$$

and

$$Q_X f(x, y) = \sum_{j=1}^m \sum_{i=1}^{\ell} \langle f, \psi_{x_i, y_j} \rangle \beta_{ij}(x, y).$$

Then for $f \in V^*(R, S, \delta)$, we have

$$\begin{aligned} \|f - S_X f\|_{L^{p,q}(\mathbb{R}^{n+1})} &= \|Tf - TQ_X f\|_{L^{p,q}(\mathbb{R}^{n+1})} \\ &\leq \|K\|_W (\|f - Q_X f\|_{L^{p,q}(C_{R,S})} + \|f\|_{L^{p,q}(C_{R,S}^c)}). \end{aligned}$$

Now for $(x, y) \in C_{R,S}$

$$\begin{aligned} |f(x, y) - Q_X f(x, y)| &\leq \sum_{j=1}^m \sum_{i=1}^{\ell} |f(x, y) - \langle f, \psi_{x_i, y_j} \rangle \beta_{ij}(x, y)| \\ &\leq M\eta^{n+1} w_{\varepsilon+\eta}(f)(x, y). \end{aligned}$$

Therefore,

$$\begin{aligned} \|f - S_X f\|_{L^{p,q}(\mathbb{R}^{n+1})} &\leq \|K\|_W (M\eta^{n+1} \|w_{\varepsilon+\eta}(K)\|_W + \delta \|K\|_W) \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} \\ &\leq (M\eta^{n+1} (\varepsilon + \eta)^\theta + \delta) \kappa_\theta^2 \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}. \end{aligned}$$

The k -th iteration can be written as

$$f_k - f_{k-1} = (I - S_X)f_0, \text{ and } f_k = S_X f + \sum_{j=1}^k (T - S_X)^j S_X f,$$

where I is an identity operator on $L^{p,q}(\mathbb{R}^{n+1})$. We define an operator $R : L^{p,q}(\mathbb{R}^{n+1}) \rightarrow L^{p,q}(\mathbb{R}^{n+1})$ as

$$R := I + \sum_{j=1}^{\infty} (T - S_X)^j.$$

If $(M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta) \kappa_\theta^2 < 1$, the operator R bounded and the operator bound is given by

$$\begin{aligned} \|Rf\|_{L^{p,q}(\mathbb{R}^{n+1})} &\leq \|f\|_{L^{p,q}(\mathbb{R}^{n+1})} + \sum_{j=1}^{\infty} \|(I - S_X)^j T f\|_{L^{p,q}(\mathbb{R}^{n+1})} \\ &\leq \left(1 + \frac{\kappa_\theta}{1 - (M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta) \kappa_\theta^2} \right) \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}. \end{aligned}$$

Therefore, R is pseudo-inverse of pre-contraction operator S_X , i.e., $RS_X f = S_X Rf = f$. This implies,

$$\begin{aligned} \|f_k - f\|_{L^{p,q}(\mathbb{R}^{n+1})} &= \left\| \sum_{j=k+1}^{\infty} (I - S_X)^j S_X f \right\|_{L^{p,q}(\mathbb{R}^{n+1})} \\ &\leq \sum_{j=k+1}^{\infty} \|I - S_X\|_{L^{p,q}(\mathbb{R}^{n+1})}^j \|S_X f\|_{L^{p,q}(\mathbb{R}^{n+1})} \\ &\leq \frac{1 + (M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2}{1 - (M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2} \\ &\quad \times \left((M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2 \right)^{k+1} \|f\|_{L^{p,q}(\mathbb{R}^{n+1})}. \end{aligned}$$

Hence, the reconstruction function f_k converges to f exponentially if the covering radius for $C_{R,S}$ associated to sample set X satisfies

$$\varepsilon < \left(\frac{\kappa_\theta^{-2} - \delta}{M\eta^{n+1}} \right)^{\frac{1}{\theta}} - \eta.$$

In other words, the equation (3.3) implies the iterative reconstruction algorithm converges with probability at least

$$\begin{aligned} &1 - \left(\frac{4}{((\kappa_\theta^{-2} - \delta)/M\eta^{n+1})^{\frac{1}{\theta}} - \eta} \right)^{n+1} R^n S \exp \\ &\quad \left(-\ell m \frac{\left(((\kappa_\theta - \delta)/M\eta^{n+1})^{\frac{1}{\theta}} - \eta \right)^{n+1}}{2^{n+1} R^n S} \right). \end{aligned}$$

This complete the proof. $\square$

Corollary 4.2. Let $\delta < \kappa_\theta^{-2}$, and $\eta < \left(\frac{\kappa_\theta^{-2} - \delta}{M} \right)^{\frac{1}{n+1}}$. If X is a discrete sample set of $C_{R,S}$ with covering radius $\varepsilon < \left(\frac{\kappa_\theta^{-2} - \delta}{M\eta^{n+1}} \right)^{\frac{1}{\theta}} - \eta$, then for any δ -concentrated function $f \in V^*(R, S, \delta)$ and

$$k + 1 > \frac{\log(\frac{1}{\delta})}{\log\left(\frac{1}{M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta}\right) - 2 \log \kappa_\theta},$$

the reconstruction function f_k defined in Theorem 4.1 is $2C\delta$ -concentrated on $C_{R,S}$, where $C = \frac{1 + (M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2}{1 - (M\eta^{n+1}(\varepsilon + \eta)^\theta + \delta)\kappa_\theta^2}$.

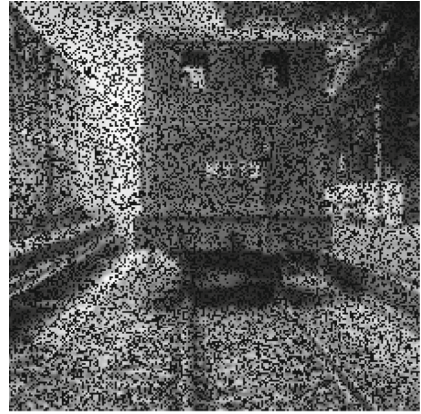
4.1. Numerical Illustration

To illustrate the proposed algorithm, we consider the function space

$$V = \left\{ \sum_{j \in \mathbb{Z}} \sum_{i \in \mathbb{Z}} c_{i,j} \phi_{i,j} : \sum_{i,j \in \mathbb{Z}} |c_{i,j}|^2 < \infty \right\}.$$



(a) Original Data



(b) Random Average Data



(c) Initial Approximation



(d) 5th Approximation

FIGURE 1. Plots of image data reconstructed from average value.

The space V is generated by $\phi_{i,j}(x,y) = \exp\left(-\frac{(x-\tau_i)^2}{3} - \frac{(y-\tau_j)^2}{3}\right)$, where $\tau_i, \tau_j \in \alpha\mathbb{Z}$ and $\alpha > 0$. The function space V is an image of an idempotent integral operator with integral kernel

$$K(x,y,s,t) = \sum_{i,j \in \mathbb{Z}} \sum_{k,l \in \mathbb{Z}} A_{i,j} B_{k,l} \phi_{i,j}(x,y) \phi_{k,l}(s,t),$$

where $A = (A_{i,j})_{i,j \in \mathbb{Z}}$ and $B = (B_{k,l})_{k,l \in \mathbb{Z}}$ are the inverse of an infinite matrix $M = (M_{i,j})_{i,j \in \mathbb{Z}}$ and $M_{i,j} = \sqrt{\frac{3\pi}{2}} \exp\left(-\frac{1}{6}(\tau_i - \tau_j)^2\right)$, see [6]. In fact, the integral kernel K satisfies $\kappa_\theta < \infty$ with $\theta = 1$.

In our numerical simulation, we consider a 8-bit image (Fig. 1a) which is represented by finite linear combination of $\{\phi_{i,j} : \tau_i, \tau_j \in \frac{25}{31}\mathbb{Z}, -62 \leq i, j \leq$

$62\}$, and the average function

$$\psi_{x_i, y_j}(x, y) = \chi_{[x_i - \eta, x_i + \eta]}(x) \chi_{[y_j - \eta, y_j + \eta]}(y),$$

where $\eta = 0.3$. From the choice of f one can verify that the function f is concentrated on $[-55, 55]^2$.

We now apply Theorem 4.1 for the function f . For our convenience, we consider the random sample points of size 250×250 chosen uniformly from the grid points $\frac{25}{56}\mathbb{Z}^2 \cap [-55, 55]^2$, so that the partition of unit function can be easily realized with support in the neighborhood of random sample points. Figure 1b denotes the corresponding random average value of the function f . Hence, using Theorem 4.1, we construct the initial approximate function (see Fig. 1c) from random average values of f , and we estimate the approximate function for the first five iterations. Figure 1d denotes the corresponding approximate function in the 5th iteration.

Figure 1a: [LCL Railway from Poland to Latvia]. April 4, 2017. BCLine. <https://www.bcline.eu/lt/lcl-gelezinkelio-linija-lenkija-latvija> Peak signal-to-noise ratio (PSNR) quantifies the quality measurement between original and reconstructed image. The higher PSNR value indicates the better quality of the reconstructed image. Let I and J denote the 225×225 order data matrix of the original image and reconstructed image, respectively. Then the corresponding mean square error (MSE) and the PSNR is defined as

$$\text{MSE} = \frac{1}{225 \times 225} \|I - J\|_{\ell^2}^2, \quad \text{PSNR} = 10 \times \log_{10} \left(\frac{MAX_I^2}{\text{MSE}} \right),$$

where MAX_I denotes the maximum possible pixel value of the image. In this case, $MAX_I = 1$. In the field of wireless transmission, the PSNR values is generally considered in the range 30 dB to 50 dB. The PSNR value of the image corresponding to the random average data of f is 15.46 dB. In Table 1, we give the mean square error MSE and PSNR values for the first five iterations.

Recently, Goyal et al. [16] introduced a reconstruction algorithm to recover the signal from their sample values at random sample points in the context of mixed Lebesgue space. In numerical simulation, we may treat the random average sample values of the signal as equivalent to random sample values and employ the sampling reconstruction algorithm proposed in [16]. In this regard, Fig. 2 illustrates a comparative analysis of changes in MSE and PSNR values at each iteration between the two methods and shows that the average sampling reconstruction scheme proposed in Theorem 4.1 performed better than the exact sampling reconstruction method. Furthermore, an analogous comparison conducted by A. Kumar and S. Sivananthan [26] within classical Lebesgue space supports our conclusion and emphasizes the significance of the average sampling method.

TABLE 1. MSE and PSNR value at n th iteration

n	0	1	2	3	4	5
MSE	4.2×10^{-4}	1.6×10^{-4}	8.1×10^{-5}	4.7×10^{-5}	2.9×10^{-5}	1.9×10^{-5}
PSNR (in dB)	33.73	38	40.89	43.26	45.28	47.07

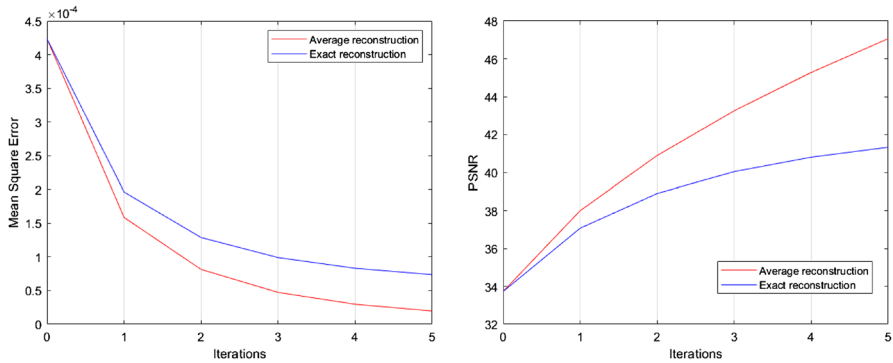


FIGURE 2. The average and classical sampling reconstruction is compared with the given average sample information of the image signal at the random sampling points. The figure on the left compares the mean square error at each iteration. The figure on the right compares the PSNR value of the image at each iteration

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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